# Data Analytics Foundations

## Module 1: Data and the data analyst role

DeepLearning.AI

# Data Analytics Foundations

Welcome to data analytics!

DeepLearning.AI

# Data Analytics Foundations

## Generative AI in this course

DeepLearning.AI

# LLMs in this course

🗓️ Demonstrates the most up-to-date capabilities as of 2025

🌲 Evergreen principles:

- How to **think about** and **use** generative AI regardless of product

🧠 Develop a mindset of iteration and experimentation

**LLM progress since launch**

- Abilities have advanced rapidly
- New features constantly being released

**Changes you should expect**

- More advanced and specialized features
- Cheaper tools
- Faster tools
- Higher quality outputs overall

# LLM options

❌ **Won't** need to purchase any additional products

❌ This course **does not** recommended any single tool

✅ Develop confidence experimenting and selecting tools

✅ See several tools throughout the modules

✅ Learn core principles to work with LLMs

# A typical day

## ☀️ Morning

- 9:00 — **Explore** new problem
- 10:00 — **Meet** data team
- 11:00 — **Work** work work!

## 🌥️ Afternoon

- 2:00 — **Create** dashboard
- 3:00 — **Present** progress
- 3:30 — **Celebrate**!
- 4:00 — **Learn** a new skill
- 6:00 — **Happy hour!**

# Data and the data analyst role

Module 1 introduction

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# Module 1 outline

**What is data analytics?**
- Multidisciplinary
- Ancient

**Spreadsheet fundamentals**
- Sort and filter data
- Write formulas
- Transform data

**Data exploration with LLMs**
- Prompting a LLM

**Time series data**
- Trends, seasonality, cyclicality
- Analytical methods

# Data and the data analyst role

Life as a data analyst

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# What’s to love about data analytics

Always discovering something new

Connects with different fields

Competitive salaries and job security

People from different backgrounds

Constantly learn new technical skills

**Great role for people who love:**

✅ Learning ✅ Problem-solving ✅ Making discoveries

# A year or two

Complete several big projects

Create a portfolio of your work

Develop industry expertise

Earn the trust of your teammates

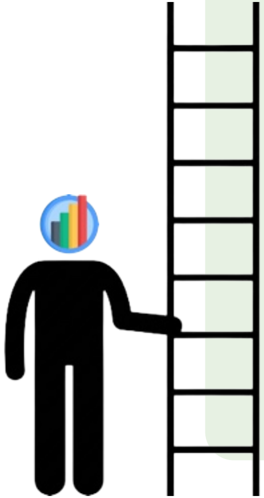
# Career of a data analyst

You may find yourself:

- Switching industries and gaining a breadth of experience

**OR**

- Developing deep expertise in a particular technical area



**Types of leadership**

Manager

Individual contributor

Educator

Mentor

**Start your own**

Business

Consulting firm

# Data and the data analyst role

## What is data analytics?

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# What is data analytics?

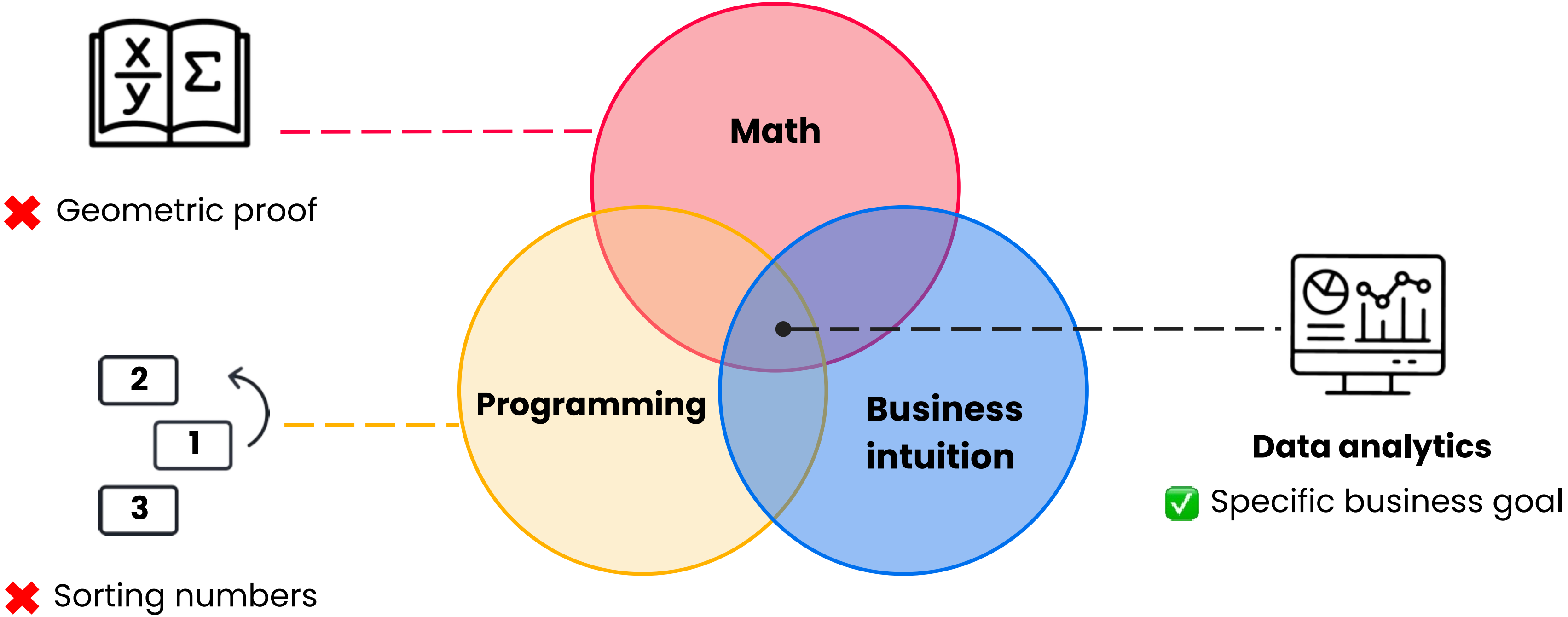
A diverse set of skills and tools that enable business to **make better decisions**.

Leveraging data to:

- ✅ Gain insights
- ✅ Support decision-making

- ❌ Relying on chance or experience alone

# What is data analytics?



# Other investigative roles

| Role | | | |
|---|---|---|---|
| Scientist | Starts with specific hypothesis | ➡️ | Analyzes data to evaluate the hypothesis |
| Detective | Gathers evidence | ➡️ | Pieces it together to understand the crime |
| Journalist | Synthesizes information | ➡️ | Creates a compelling narrative |

# Data analytics vs. data analysis

## Data analytics

| | Data analytics |
|---|---|
| Scope: | • Broader scope<br>• Real-time and predictive modeling |
| Techniques: | • More sophisticated techniques<br>  ○ Advanced programming<br>  ○ Visualization software<br>  ○ Big data<br>• More complex, iterative process |
| Business integration: | • Deeply integrated into systems<br>• Aims to:<br>  ○ Forecast trends<br>  ○ Guide decisions<br>  ○ Interpret past data |

## Data analysis

- Tracking budget over time in a spreadsheet

## Data analytics

- Use statistical modeling techniques on large datasets from multiple sources
- Create visualizations to identify the most promising revenue streams
- Integrate insights into a real-time decision-making system

## What's not new 🗿

- Statisticians, scientists, and engineers analyzing data
- Data visualization

🥇 1150 BC, Turin Papyrus Map

## What's new ✨

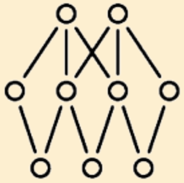
- Explosion of data
- More powerful tools

**96,000**
Taylor Swift fans

**615,000,000**
active users

# Distinction between data roles

**Data scientists**



Use **sophisticated** modeling techniques

**Data analysts**

Use **foundational** statistical techniques and data visualization

**Business intelligence analysts**

Use **commercial** software like spreadsheets, with lighter emphasis on programming

Your work may fall under several of these domains!



Sean Barnes

# Data and the data analyst role

Evidence-based decision-making

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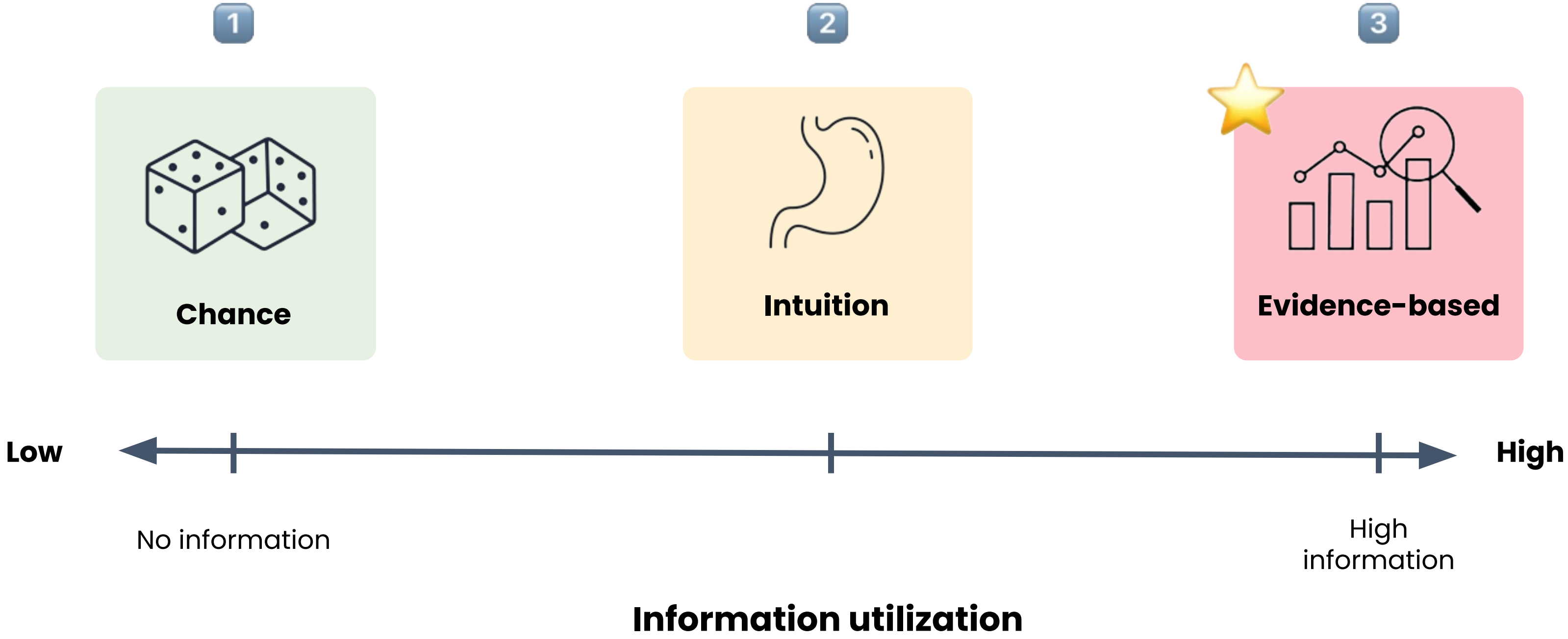
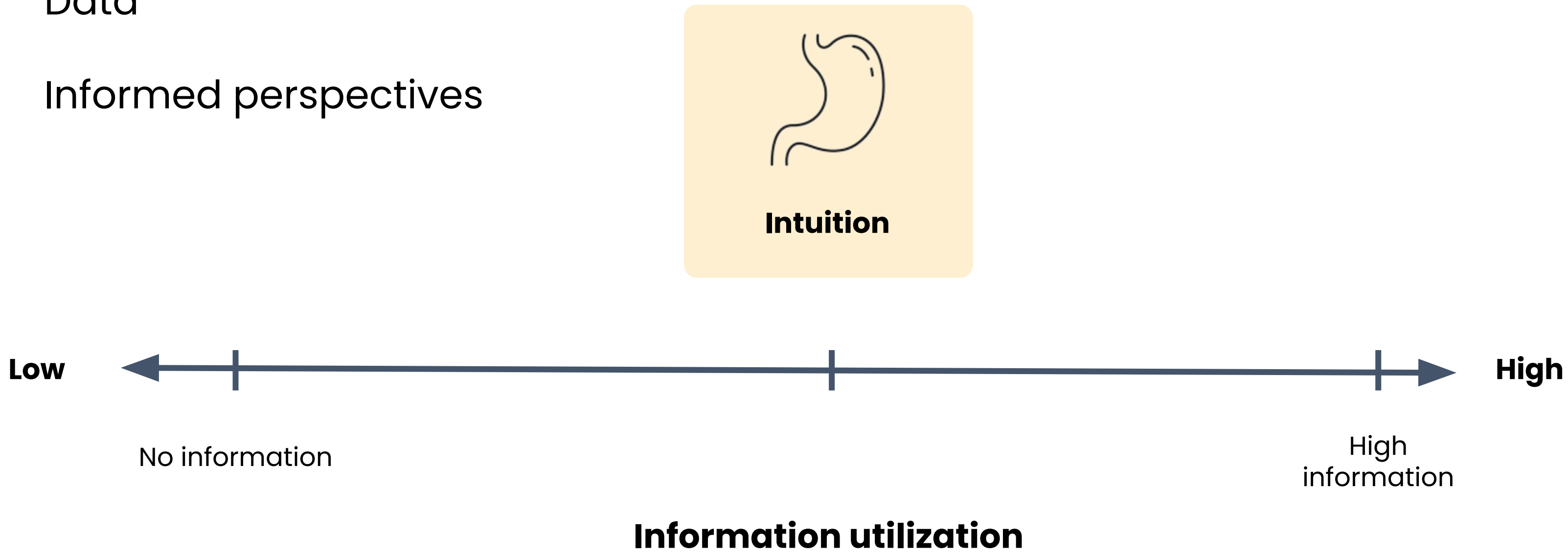
# Decision-making methods



# Intuition

Other information to consider:

- 📊 Data
- 💡 Informed perspectives

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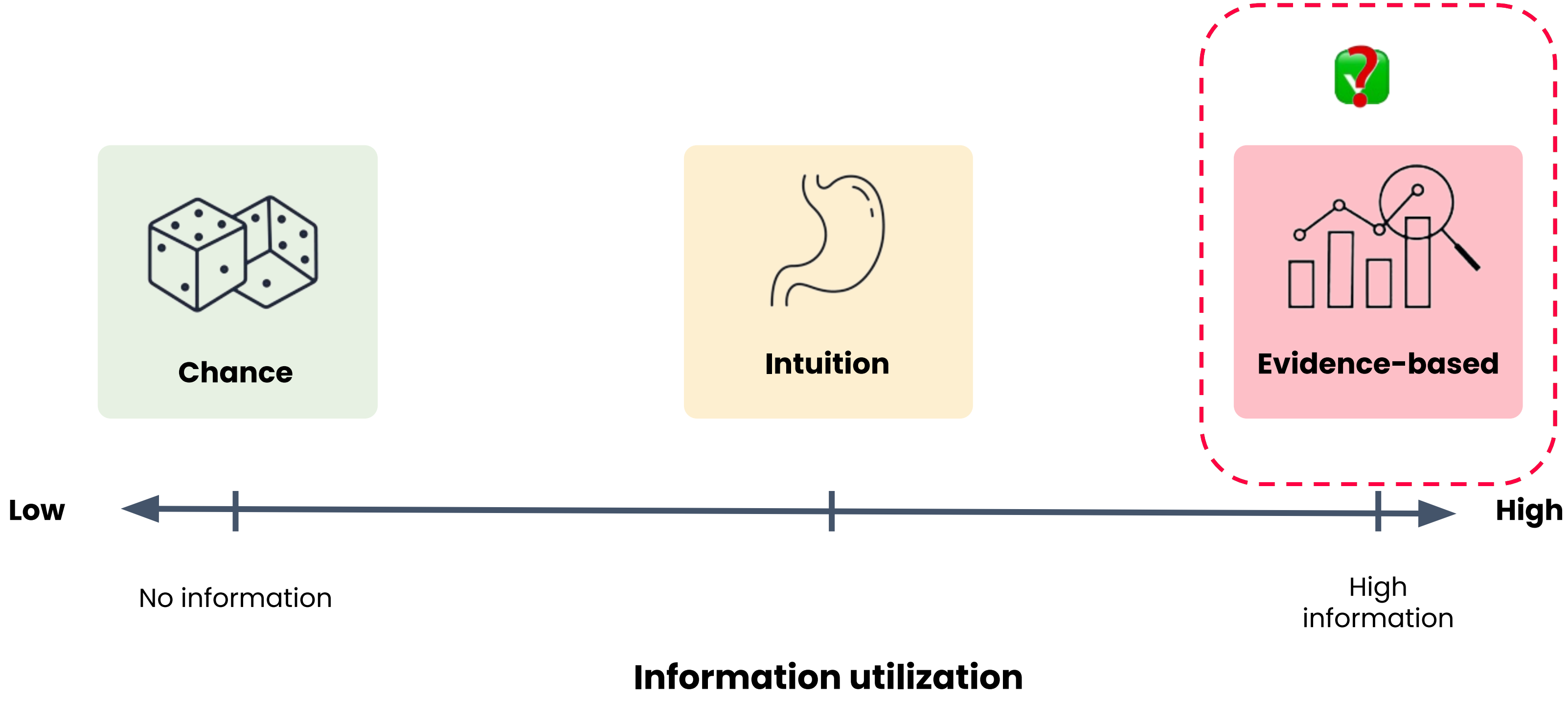
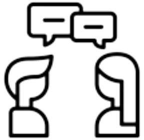
# Decision-making methods



# Decision-making methods

Which university should I choose?

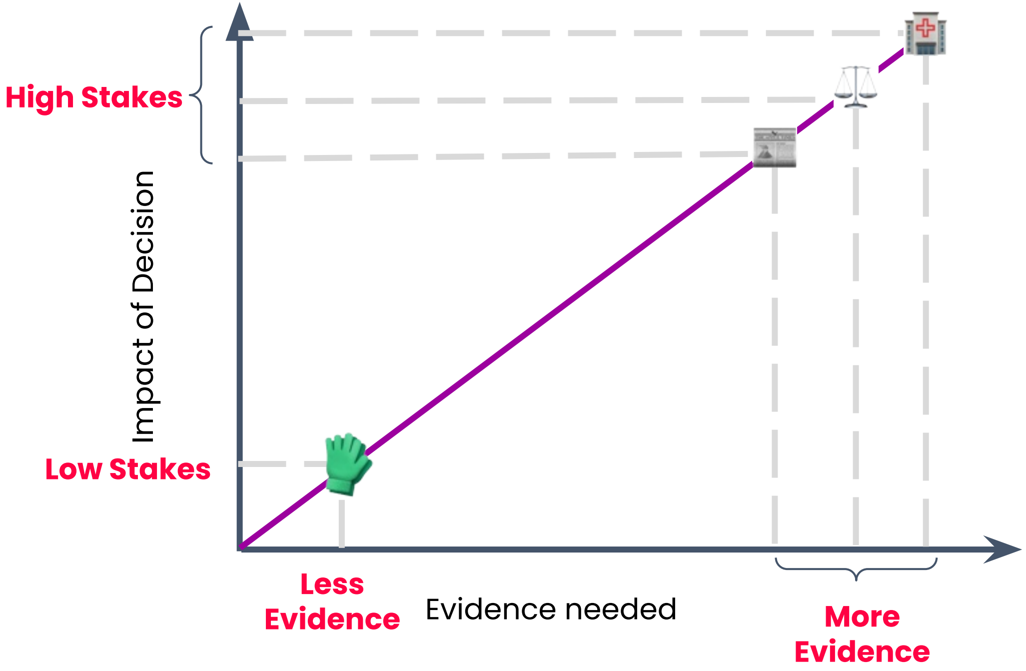
Is it smart for me to move to a new city?



How should I invest my savings?

# Effort vs. impact in decision-making

- Level of effort needed to accumulate evidence is **proportional** to the impact of the decision.



# When to trust your gut

Intuition helps you:

- ✅ Make quick, low stakes decisions
- ✅ Avoid searching through data without any idea of what you might find

**Evidence-based**

**Low** **High**

Limited historical data

Intuition & data

**Information utilization**

# Example


✅ Open a new store in your city

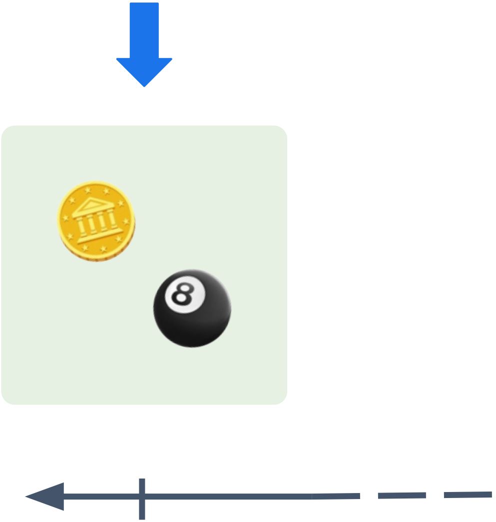
✅ Improve employee benefits

✅ Offer greater variety of fish

Options to increase revenue:

- [ ] Adding more reptiles
- [ ] Staying open 2 extra hours per day
- [ ] Raising prices on animal feed 🤔

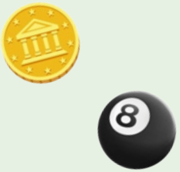
# Choosing a decision-making method



Low

No information

# Choosing a decision-making method



**Advice from grandma:**

**“Offer more reptile varieties”**

**Low**

No information

Limited historical data

# Choosing a decision-making method

**Advice from grandma:**

**“Offer more reptile varieties”**

- **Define problem and outcome**
- **Gather data on reptile varieties**
- **Gather information**
- **Experiment with later store hours**
- **Synthesize information**
- **Conduct surveys on higher prices**

**Low** ← → **High**

No information

Limited historical data

Intuition + data

# Data and the data analyst role

A history of data analytics

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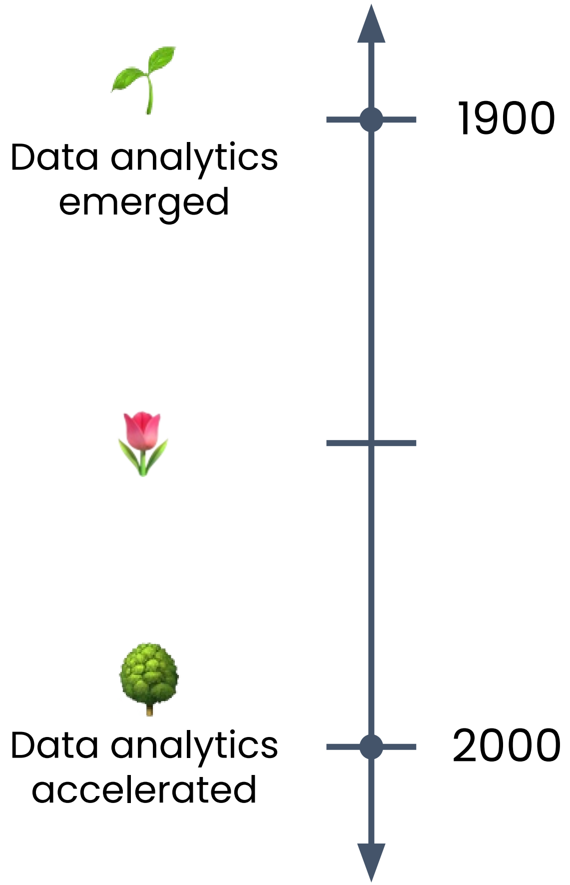
# The history of data analytics

## 🔄 Continuous improvement

- ✨ Process of trying to improve products, services, and processes
- ✅ Adapt to the evolving competitive landscape around you
- ❌ Staying the same rarely leads to long-term survival



# Data analytics in the military

**1940s**: Military **operations research** during WWII.

🎯 **Problem**:

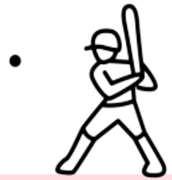
Placing radar equipment in the most optimal setup to protect Britain

❌ Allies didn't have time to experiment

**Outcome:** Used data

- Geographic information
- Enemy aircraft flight patterns
- Radar range
- Signal strength

# Data analytics in baseball



## Intuition-based scouting

### "Eye Test"

- Looks good?
- Athletic?
- Seems to have potential?

### Optimized player recruitment

- More nuanced analyses
- Increase in computing power
- Lessened emphasis on intuition

1970s

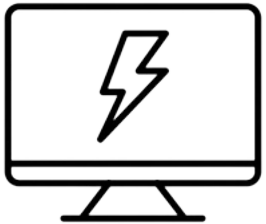
# Data analytics in baseball


- Statistician Bill James began publishing innovative new statistics
- Detailed evidence about how much a player contributes to winning

- Oakland Athletics, a team with a small budget, was able to recruit undervalued players
- Gained an advantage over the richest teams
- Their players didn't look typical, but won more often

# Key trends

Explosion of available data



Rapid advances in computing power

Growing interest in evidence-based decision-making

# Data analytics is everywhere

## Tech


- Recommend products
- Optimize user experiences

## Retail

- Manage inventory and pricing

## Healthcare

- Improve patient outcomes
- Reduce costs

# Data and the data analyst role

Modern industry use cases

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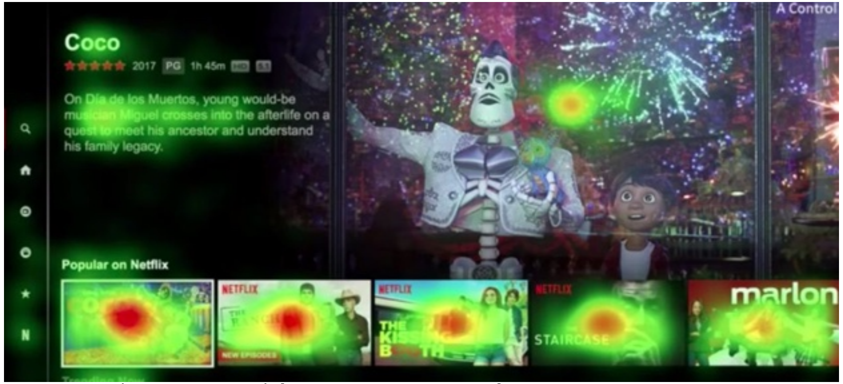
## Modern streaming companies




Product Peas. (2018, November 5). [Eye tracking results image]. In Case study 2— Zach Schendel, Director of UX Research at Netflix. Medium.

- ✅ Collect detailed data about **every action** taken by **every user** on their platform
- 🏔 **Challenge**: How to wrangle data
- 💡 **Decision**: Who to recommend shows to?
- 💡 **Decision**: How to make people like GOT?

## Old school TV ratings

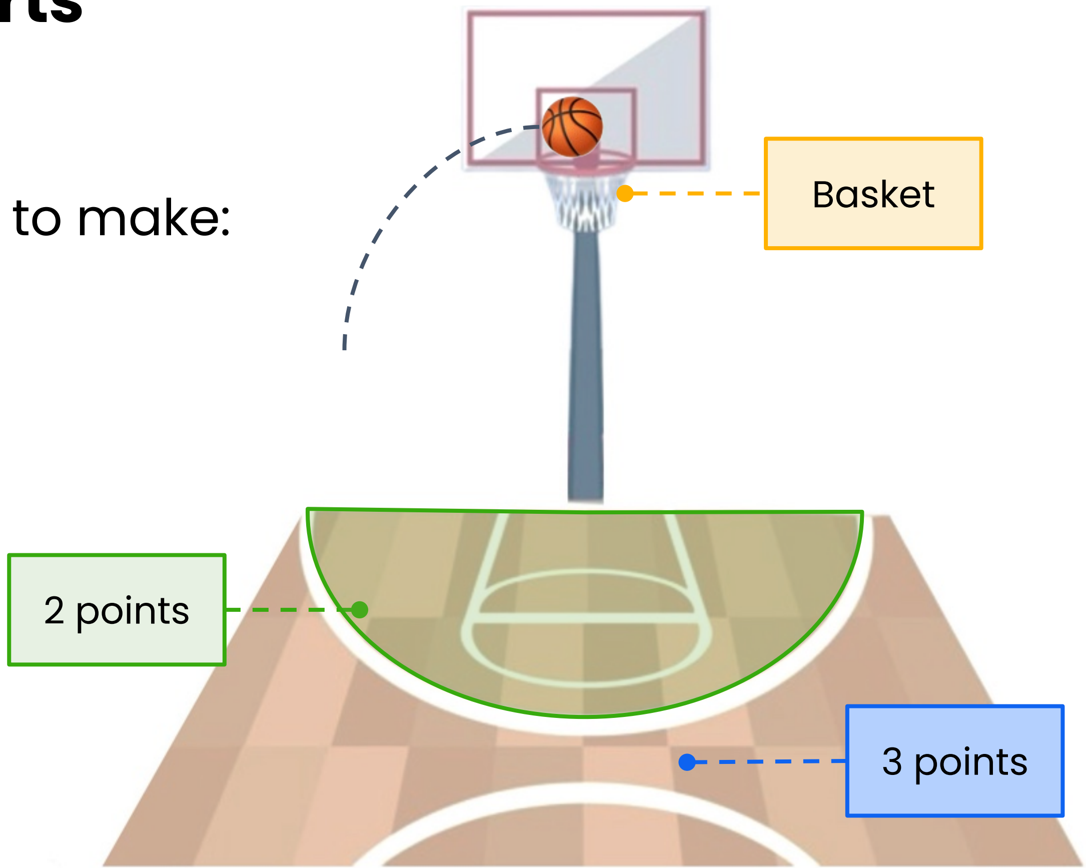

Nielsen. (2020). Nielsen People Meter [Image]. In J. Lafayette, Nielsen readies big data metrics for TV advertising. Broadcasting & Cable.

- ✅ Measured viewing habits via devices in a **sample** of households
- 🏔 **Challenge:** Recording habits of mostly older TV-watchers
- 🏔 **Challenge:** Collecting incorrect data from households with multiple TVs

# Data analytics in sports

Key decision a player has to make:

- When to try scoring



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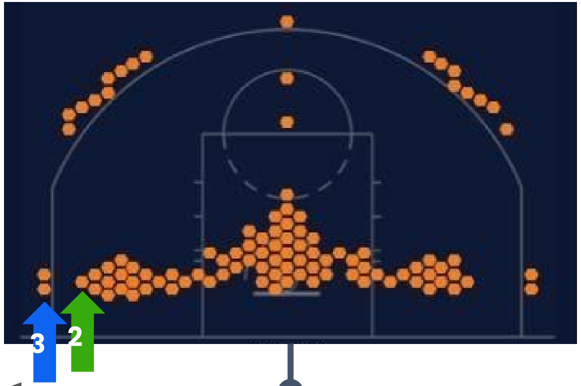
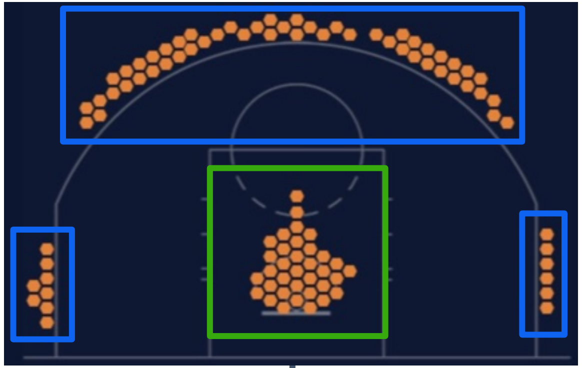
# Data analytics in sports




Data analytics

- Maximize shots with the **highest expected points**
- Take the easiest shots of each point value



1997

2019

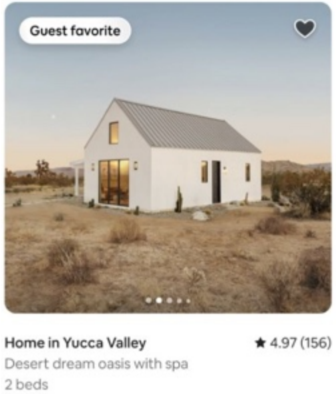
LongTermMetabolite. (c. 2020). [OC] Most frequent NBA shot locations [Reddit post]. r/dataisbeautiful. https://www.reddit.com/r/dataisbeautiful/comments/h94umw/

# Product design

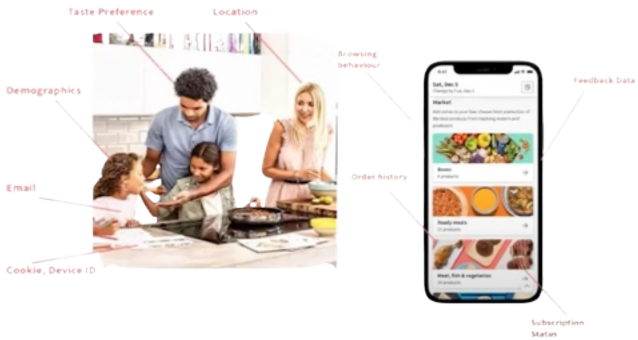
STITCH FIX


Withers, B. (2018, July 14). Data that matters [Image]. In Is This Stitch Fix's Secret Weapon? The Motley Fool.

airbnb




HELLO FRESH




Inside HelloFresh. (2022, January 11). Webinar: Data, Our Main Ingredient at HelloFresh [Video]. YouTube.

**Using data to:** 👍🏽 Power recommendation systems | 🧪 Guide experimentation

# Product design

HELLO FRESH

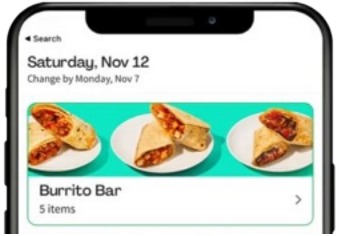
### Customer Data

- Demographics
- Location
- Taste profile

### Recipe Data


- Ingredients
- Availability
- Image of recipe
- Recipe text

### Transaction Data




- Order history
- Feedback
- Browsing behavior

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**Use data to make decisions about:**

🥑 Which products to recommend

📅 When to offer discounts

🧑‍🍳 How to choose recipes

# Education and government

**Data**

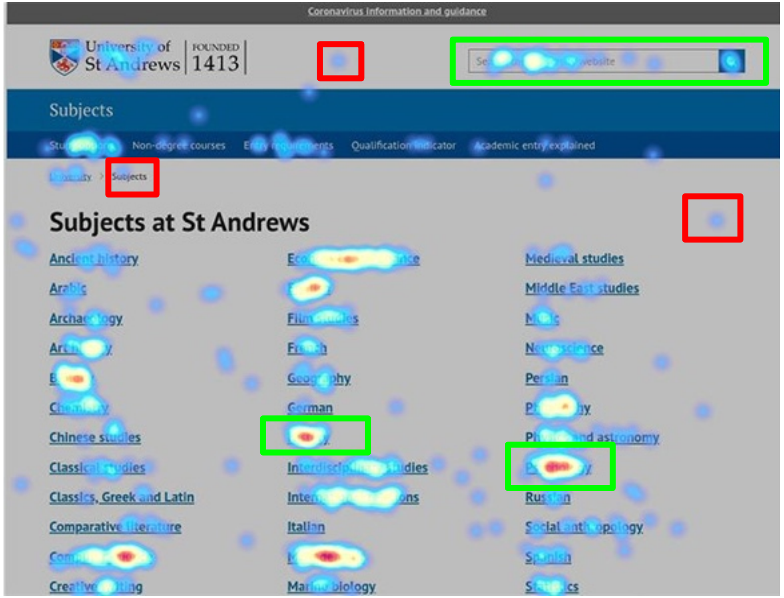
- Users are clicking popular subjects like history and psychology
- Users are clicking the search bar

**Analysis**

- Users attempted to click non-interactive page elements
- Lower parts of the page had a sharp dropoff in activity

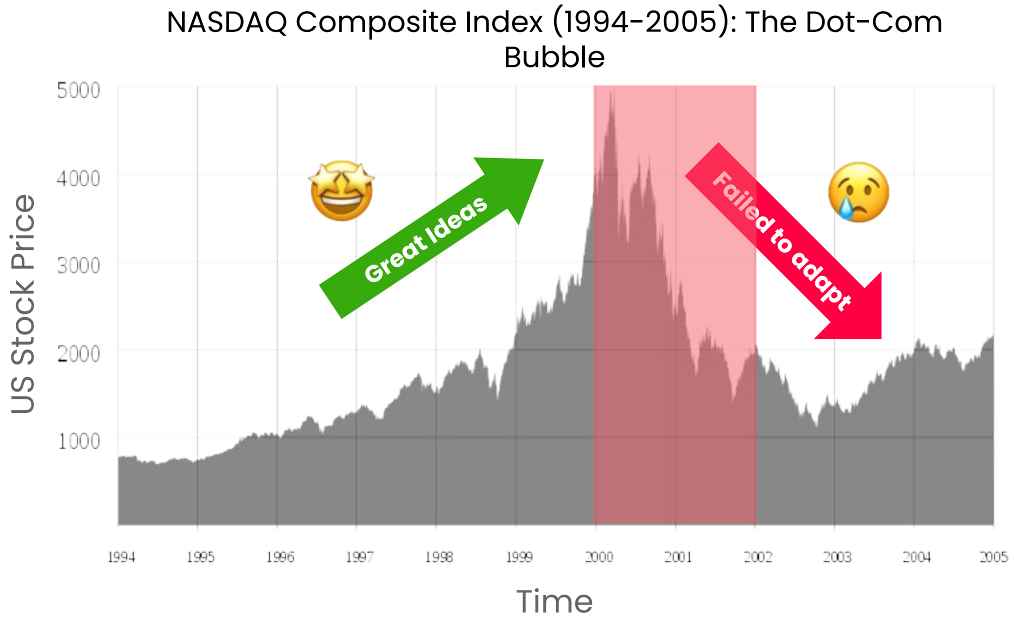
**Changes**

- Clarifying which elements are clickable
- Reprioritizing information towards the top of the page




**Use analytics to:** 📓 Improve access to information | 🧑‍💻 Improve user experience

# Continuous improvement




2000-2002 dot com crash

pets.com

❌ Struggled to **compete** with brick & mortar stores

✅ Some companies survived by **continuously reinventing** themselves

Google amazon Meta NETFLIX



Sean Barnes

# Continuous improvement

- Not every company needs to reinvent core product
- Focus on doing **one core thing** really well
- Use data analytics to **optimize** logistics or marketing
- Core product **remains largely the same**

💡 Data can help a business improve in whatever way matches its goals.


**Focus**: Producing sweet, caffeinated beverages


**Focus**: Reliably delivering packages

# Data and the data analyst role

Defining data

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# What is data?

- Information to help you make a decision
- Data gets generated just about anywhere:
  - 💡 Whether you realize it
  - 📝 Whether you record it

Numbers

Text

Images

Sounds

Social media videos

Voice recordings

Profits from last year

**Information:** how fresh the tea leaves are

**Information:** whether it's dawn or dusk

Record it to analyze it

# Tracking the sun over the years

## Ancient Times

Stonehenge

## Modern Times

Satellite imagery
& digital calendars

# Data across industries

**Sports ⚽**

Players

Positions

In-game statistics

**Retail 🛍️**

Sales

Customer behavior

**Healthcare 🏥**

Medical images

Handwritten doctors' notes

**Social media 📱**

Ad views

User interactions

**Most industries**

Payroll

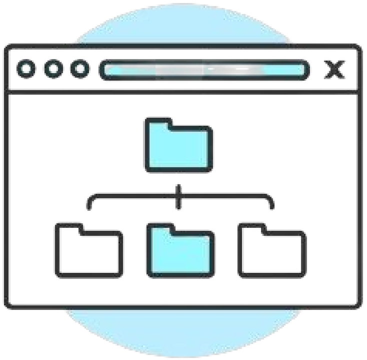
Website traffic

Electricity costs

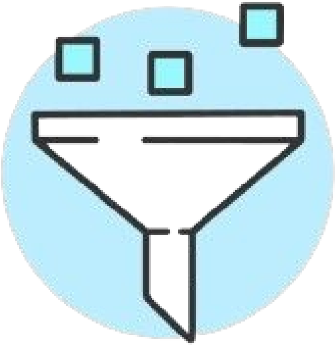
Bank balances

Legal documents

# Purpose-driven data collection



Collect data that serves a purpose



Filter through available info

Decide what's most relevant

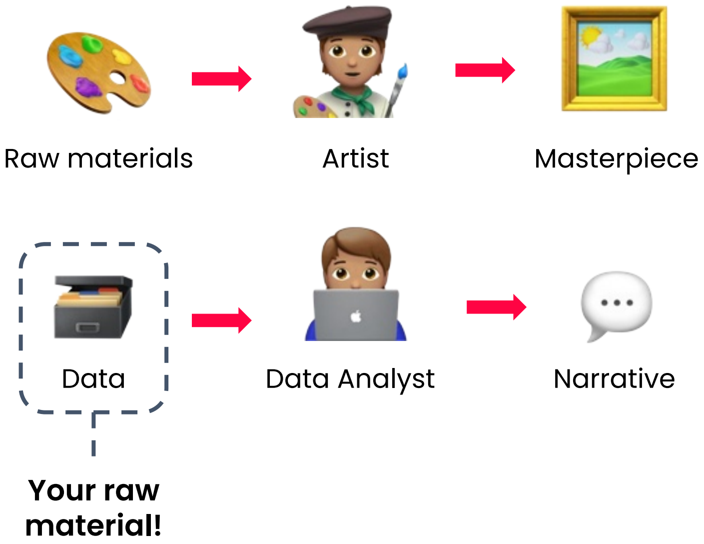
Remember: Data is information that can help you **make a decision**.

# Bringing your unique perspective

**You**
Data Analyst

Use your perspective to:

- Interpret data
- Find patterns and insights
- Tell a story



# Data and the data analyst role

Unstructured data

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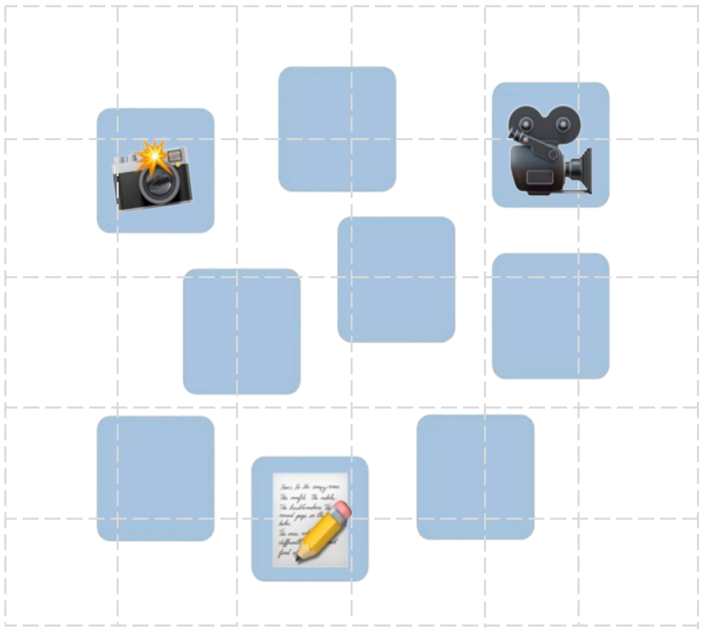
# Unstructured data

- Doesn't fit neatly into rows or columns
- Human types of information
- Feel natural



## Unstructured data

*March 1, 2027*

Today, there were 4 people in our running group.

We ran 3.2 miles in 29 mins and 32 secs.

Afterwards, I focused my strength training on my core.

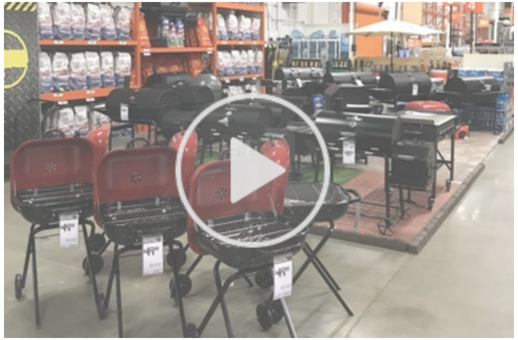
More human

## Structured data

| Day | People | Miles run | Time | Strength focus |
|---|---|---|---|---|
| 3/1/27 | 4 | 3.2 | 29:32 | Core |

More rigid, but better for analysis

## Unstructured data

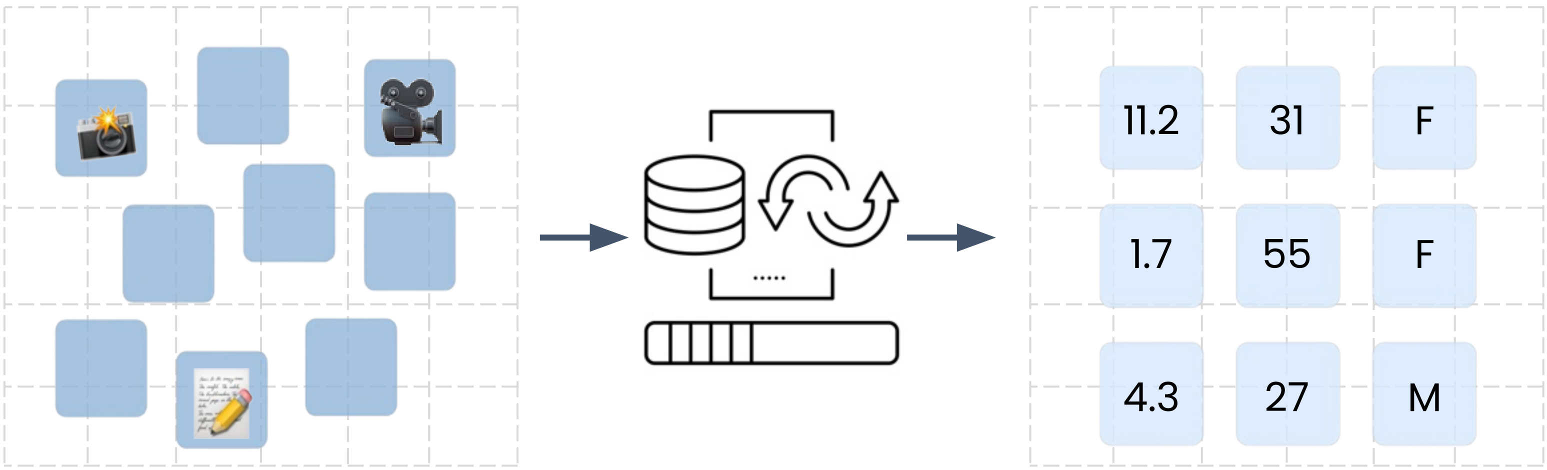
## Structured data

| Sushi item | Ingredients |
|---|---|
| California roll | Crab, avocado, cucumber |
| Spicy tuna roll | Tuna, mayo, chili |
| Veggie roll | Avocado, cucumber, carrot |

| Name | Price | Warranty | Rating |
|---|---|---|---|
| Grill A | $299 | 1 year | ★★★★☆ |
| Grill B | $399 | 2 years | ★★★☆☆ |
| Grill C | $499 | 3 years | ★★★★★ |

# Collection and processing

**Unstructured data** | **Structured data**




⚠️ Requires more steps before it becomes useful

✅ Most analytics happens on structured data

# Customer reviews


**You**

Data Analyst

**Write a review**

| | |
|---|---|
| Fun | ★★ |
| Accessibility | ★★★★ |
| Value | ★★★★★ |

Yeah the amount of bedbugs was overwhelming, but the coffee was good!

**Structured**

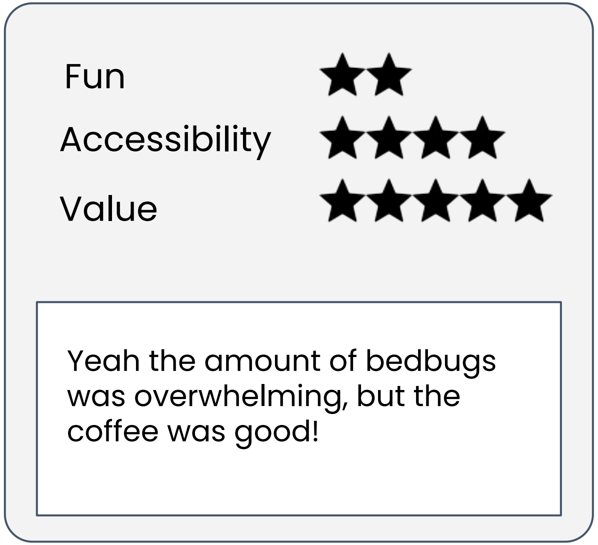
- Analyze directly
- May not be as insightful

**Unstructured**

Needs to be processed

1. Translate to English
2. Chunk for storage
3. Classify by tone

# Customer reviews




- Contain a **mix** of structured and unstructured information
- Common to store comments **along with** the structured data

| Fun | Accessibility | Value | Review |
|---|---|---|---|
| 2 | 4 | 5 | Yeah the amount of bedbugs was overwhelming, but the coffee was good! |

- Still unstructured
- Lacks inherent organization

# Data and the data analyst role

Structured data

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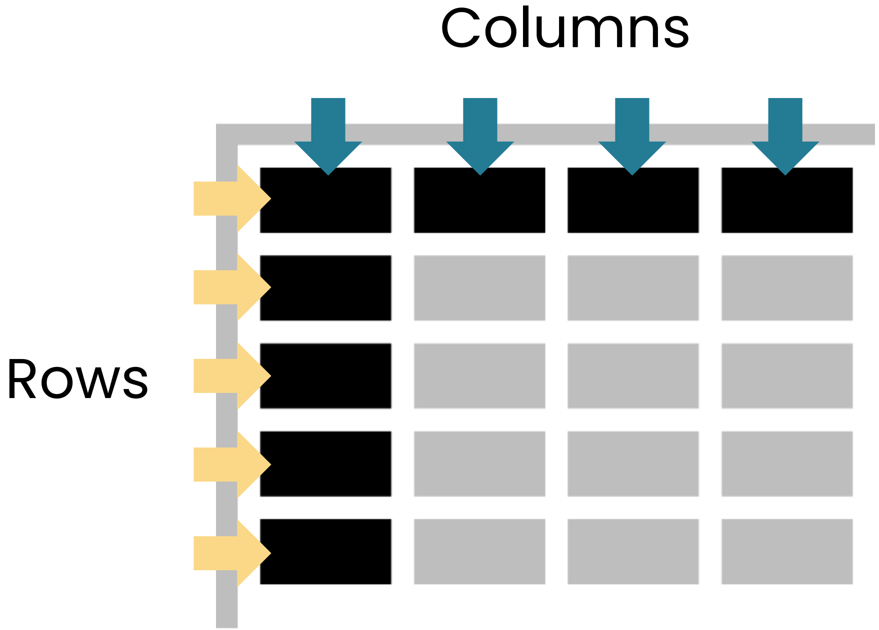
# Structured data

- Standardized format that computers can easily interpret
- Organize data neatly into rows or columns



# Tracking workouts

| Day | People | Miles run | Time | Strength focus |
|---|---|---|---|---|
| 3/1/27 | 4 | 3.2 | 29:32 | Core |
| 3/5/27 | 0 | 0 | 0:00 | Legs |
| 3/8/27 | 2 | 1.1 | 9:20 | Back |

Information about each exercise

Never negative

Will always have a time

One of few options:

- ❑ Core
- ❑ Legs
- ❑ Back
- ❑ Arms

# Data types

**Numerical Data:**

| Discrete | Continuous |
| --- | --- |
| 4 | 3.2 |
| whole numbers | fractional amounts |

**Categorical Data:**

- Separate rows into distinct groups
- Finite number of distinct groups
  - ❌ Free text

**As text**

| Strength focus |
| --- |
| Core |
| Legs |
| Back |

**Numerically**

| Strength focus |
| --- |
| 1 |
| 2 |
| 3 |

# Time series and cross-sectional data

### Time series

- Tracks **changes** over time

| Day | People | Miles run | Time | Strength focus |
|---|---|---|---|---|
| 3/1/27 | 4 | 3.2 | 29:32 | Core |
| 3/5/27 | 0 | 0 | 0:00 | Legs |
| 3/8/27 | 2 | 1.1 | 9:20 | Back |

### Cross-sectional data

- Captures a **snapshot** in time

ruggable

2,217 posts 1.3M followers 405 following

Unstructured

Perfection is overrated.
🌎 Available in 🇺🇸 🇨🇦 🇩🇪 🇬🇧 🇦🇺 🇦🇹 🇫🇷 🇳🇱 🇰🇷

| Name | Posts | Followers | Following | Bio |
|---|---|---|---|---|
| ruggable | 2217 | 13000000 | 405 | Perfection is overrated. 🌎 Available in 🇺🇸🇨🇦🇩🇪🇬🇧🇦🇺🇦🇹🇫🇷🇷🇺🇰🇷 |

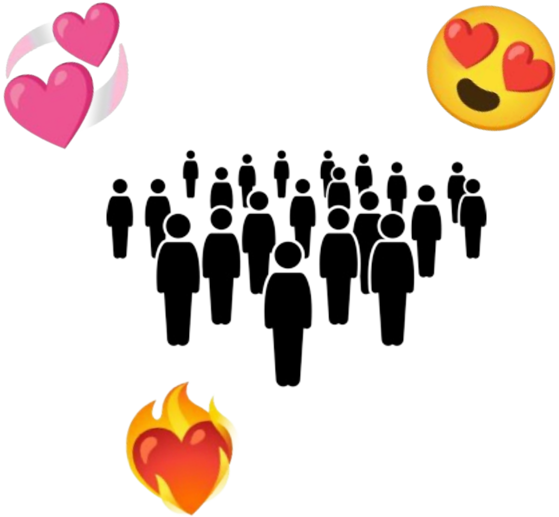
✅ Consistent format

# Structured and unstructured data

Read a book

Watch a movie

Enjoy a song



# Structured and unstructured data

Read a book

Watch a movie

Enjoy a song

✅ Generative AI is making progress on unstructured data

⛔ Non-AI techniques work best on structured data

# Text embeddings

## Unstructured data

Today there were 4 people
in our running group.

## Structured data

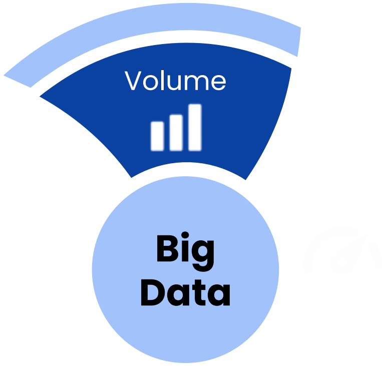
[0.85, 0.73, 0.01, ... ]
[0.08, 0.44, 0.09, ... ]
[0.18, 0.31, 0.93, ... ]

| King | - | Man | + | Woman | = | Queen |
|---|---|---|---|---|---|---|
| [0.60, ... ] | | [0.10, ... ] | | [0.20, ... ] | | [0.70, ... ] |

# Data and the data analyst role

Big data

# The v's of big data




## 1 Volume

- Data sets today are large
- Pose challenges in storage and computation
- Requires serious computational power

amazon

**Volume of orders:**
12 – 19 million per day

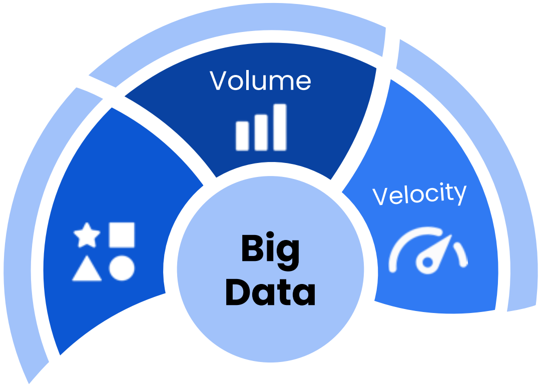
# The v’s of big data


## 2 Variety

- 21st century → explosion of **unstructured data**
- Coincides with the rise of the internet and social media

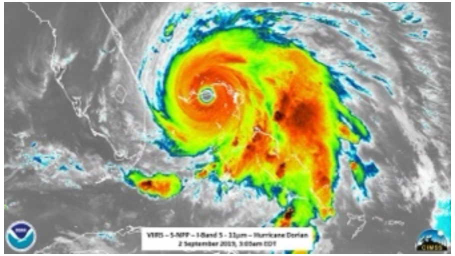

📸Add a photo
📍Check in at a location
🏷️Tag people
💸Start a fundraiser
🫠Choose a feeling
🎥Stream live video

# The v's of big data




## 3 Velocity

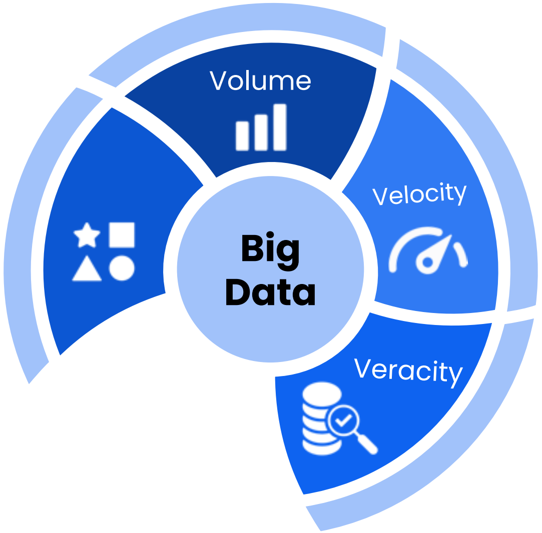
- Speed at which data is being **generated**



- Sensors and satellites collect data every second
- People in the storm's path need data quickly


- First 6 months → 100k+ views/day
- Velocity of data had consequences for content moderation

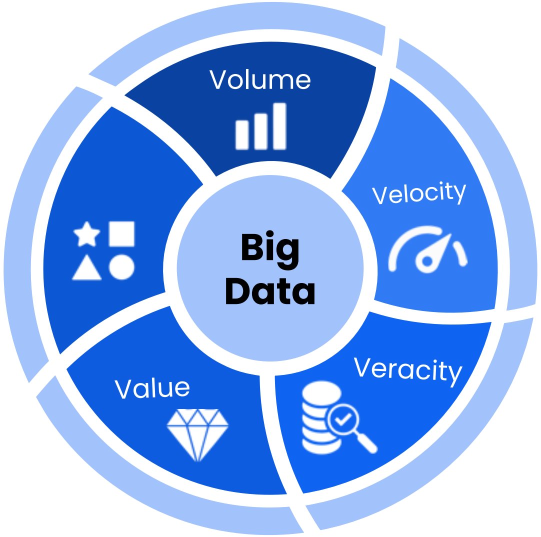
# The v's of big data




## 4 Veracity

- **Quality** of the data
- Crucial especially as volume, variety, and velocity increase
- Garbage in, garbage out

# The v's of big data




## 5 Value

- Data is worth analyzing if it provides some benefit
- Engagement data → → → recommendations
- Without data, everyone would get the same recommendations


# Small data

Small data sets produce valuable insights.

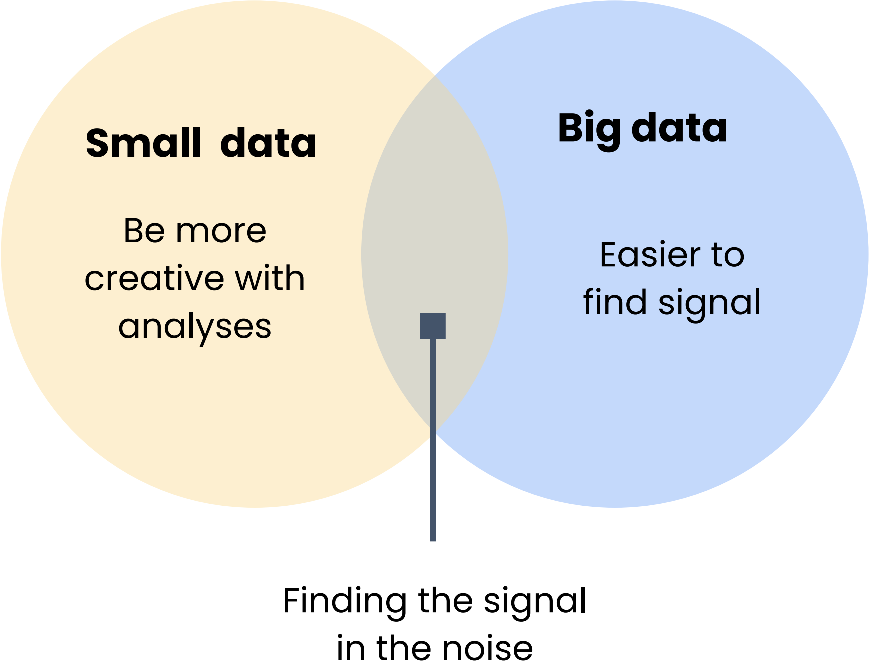

## Hospital data size

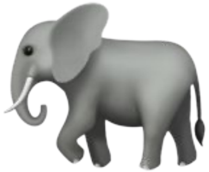
- About 6,000 hospitals
- Each hospital → a few 1000 patients per year
- Intensive care → 24 beds

✅ Feasible to analyze on your laptop

✅ Valuable for improving patient outcomes

# Challenges of big and small data



# Data and the data analyst role

Data ecosystems

DeepLearning.AI

# The data ecosystem

**Collection**

Gathering data

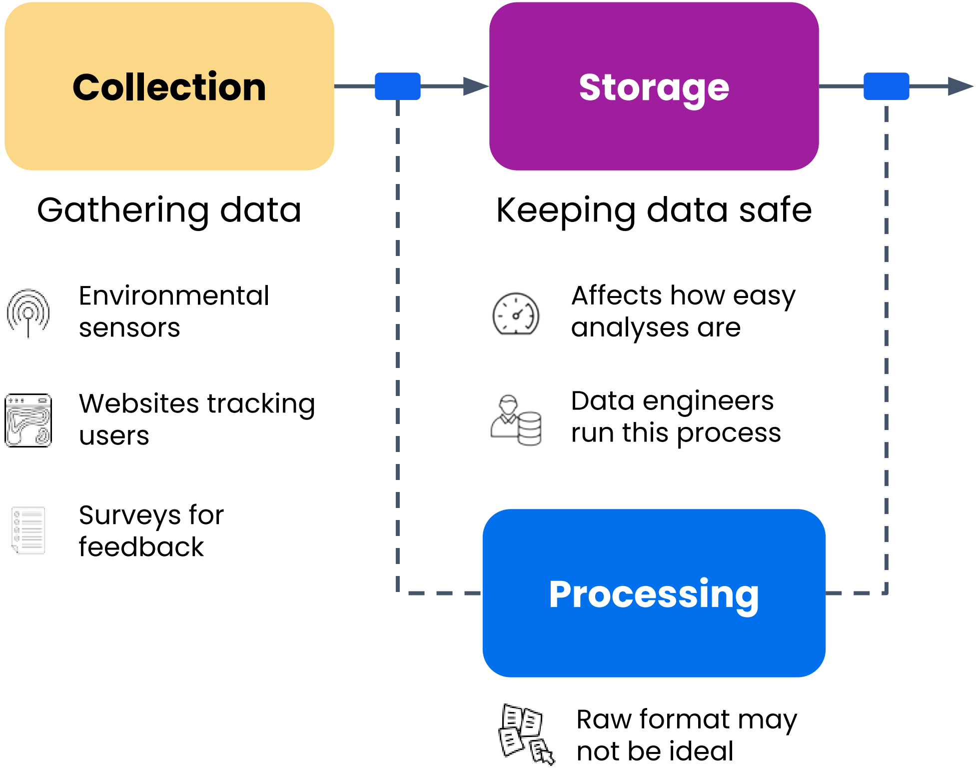
- Environmental sensors
- Websites tracking users
- Surveys for feedback

# The data ecosystem

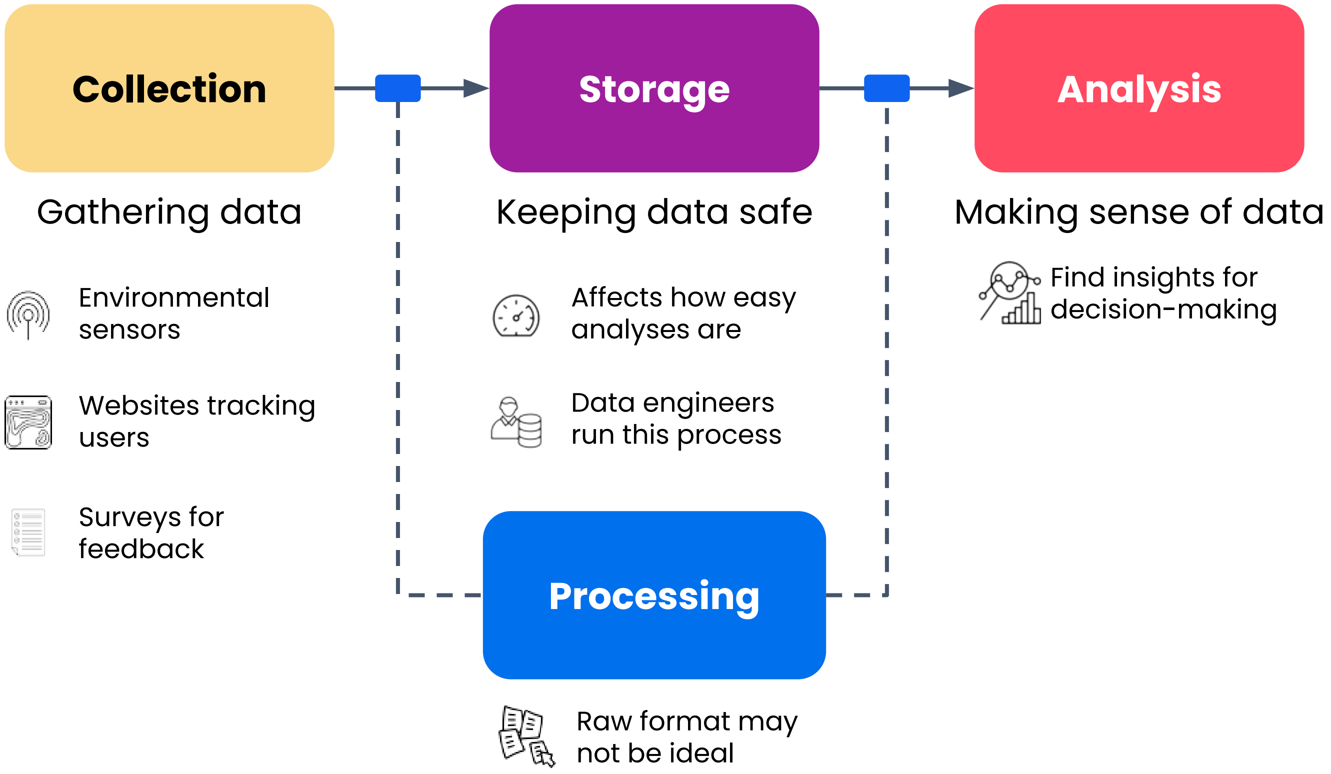
**Collection** → **Storage**

| Collection | Storage |
|---|---|
| Gathering data | Keeping data safe |
| Environmental sensors | Affects how easy analyses are |
| Websites tracking users | Data engineers run this process |
| Surveys for feedback | |

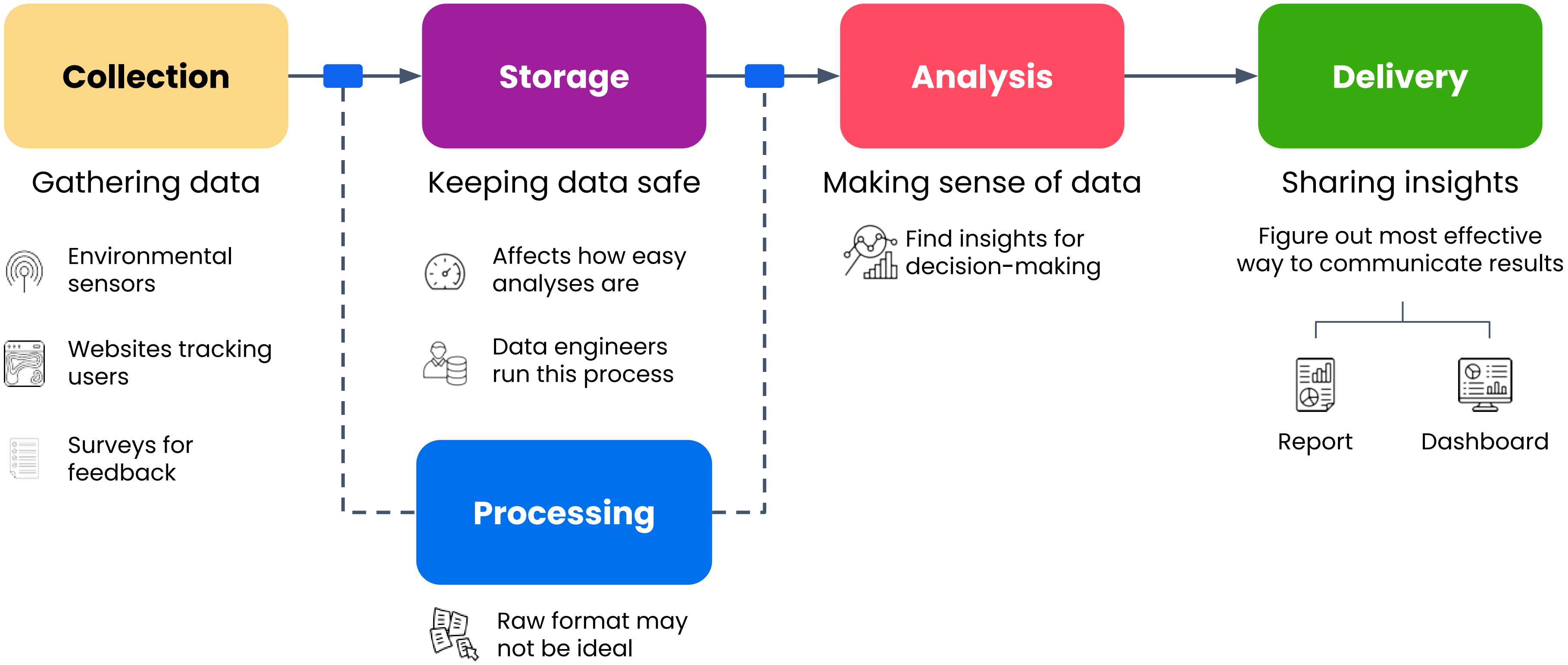
# The data ecosystem



# The data ecosystem



# The data ecosystem



# The data ecosystem

Data engineers → → Stakeholders

**Collection** → **Storage** → **Analysis** → **Delivery**

**Processing**

## Collection

Gathering data

- Environmental sensors
- Websites tracking users
- Surveys for feedback

## Storage

Keeping data safe

- Affects how easy analyses are
- Data engineers run this process

## Processing

- Raw format may not be ideal

## Analysis

Making sense of data

- Find insights for decision-making

## Delivery

Sharing insights

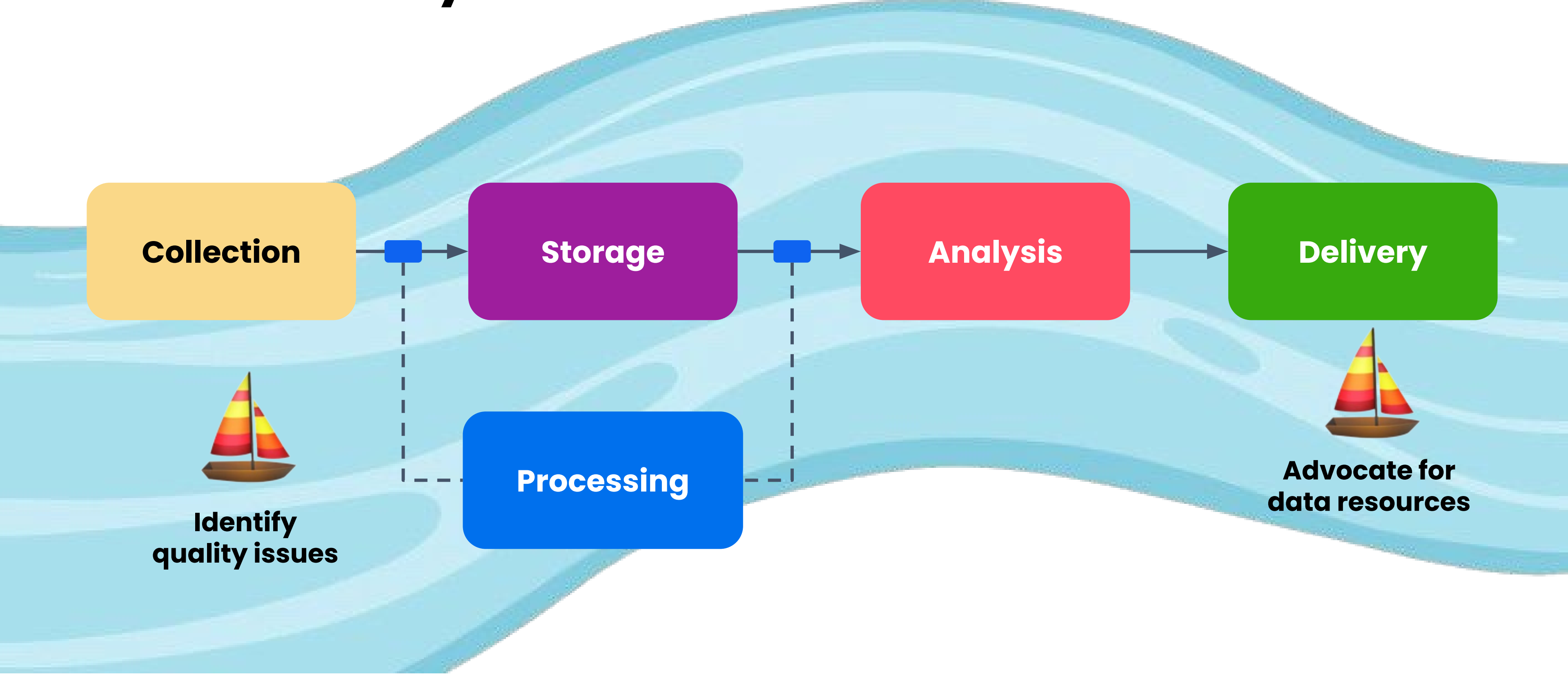
Figure out most effective way to communicate results

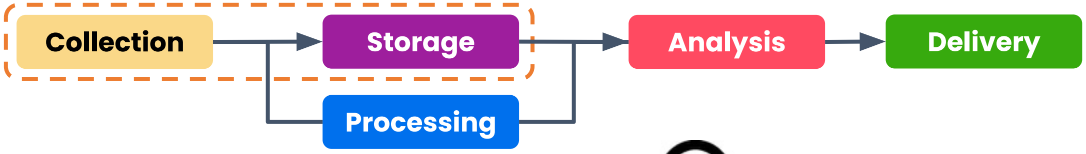
- Report
- Dashboard

Others will depend on your work:

- Users who generate the data
- Business stakeholders

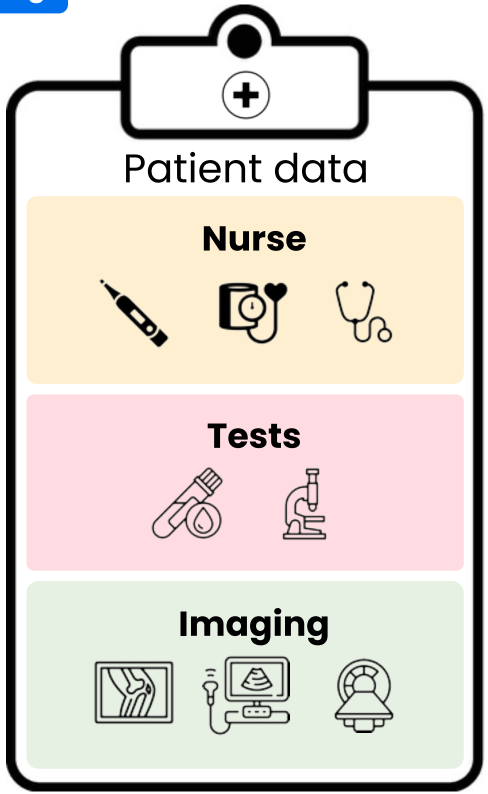
# The data ecosystem



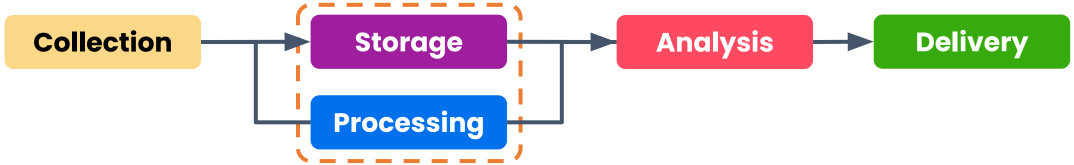


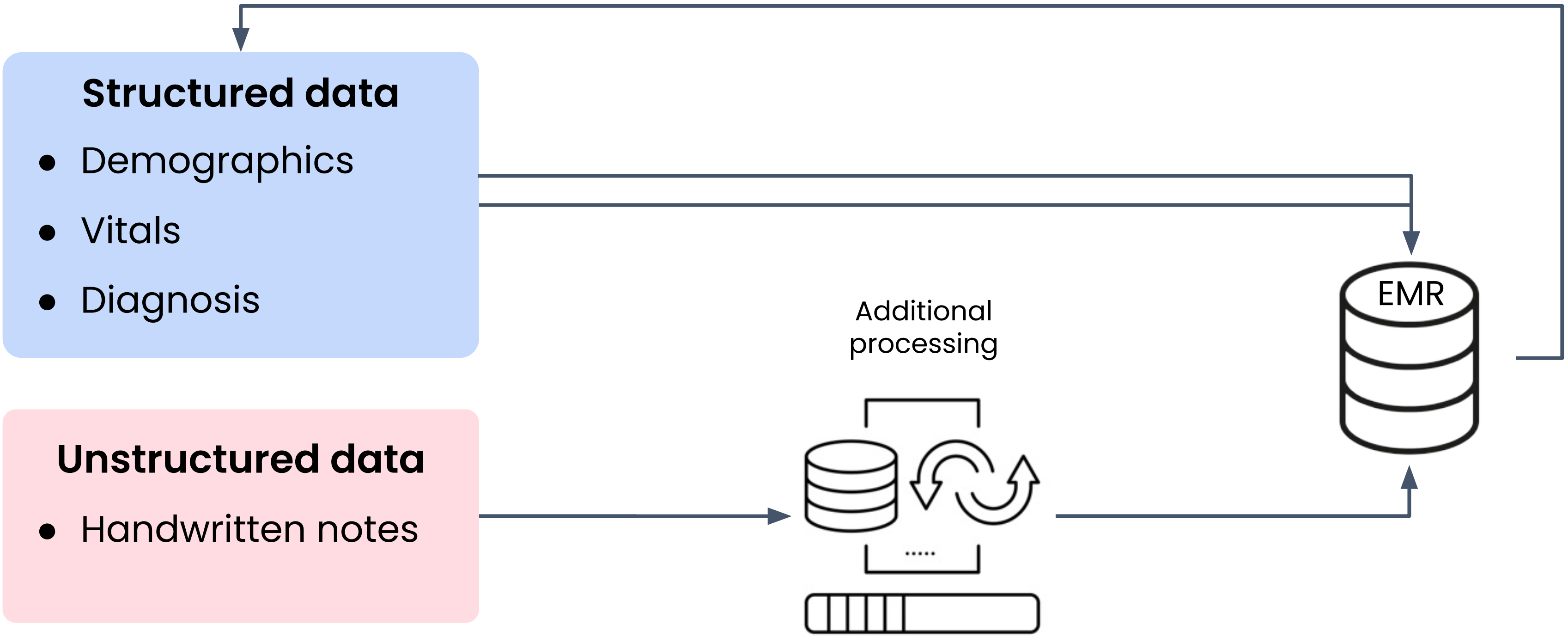

# Hospital example

- Hospital admits patients with severe unexplained symptoms
- Collected, interpreted, and stored in an **electronic medical record** (EMR)
- Attached to:
  - Patient's identity
  - When collected
  - By whom

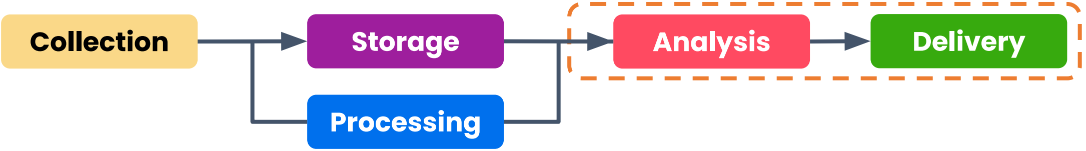


# Hospital example



# Hospital example




**You**

Data Analyst

Analyze patterns across thousands of records to:

Surface insights that help doctors make better decisions

Find a treatment protocol that leads to better outcomes

Highlight risk factors that doctors are overlooking

# Data and the data analyst role

Collaborators outside your data team

DeepLearning.AI

# Business stakeholders

**Their role**:

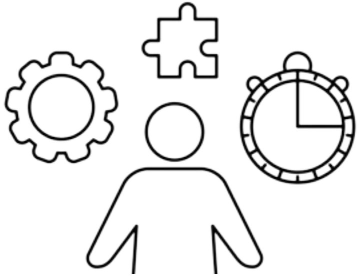
- Make decisions based on insights

**Your role**:

- Understand their problems
- Provide them with insights to make informed decisions

| Stakeholder | Goal |
|---|---|
| Restaurant owner | Optimize menu |
| Store manager | Track inventory |
| Policymaker | Evaluate new bill |

# Product managers

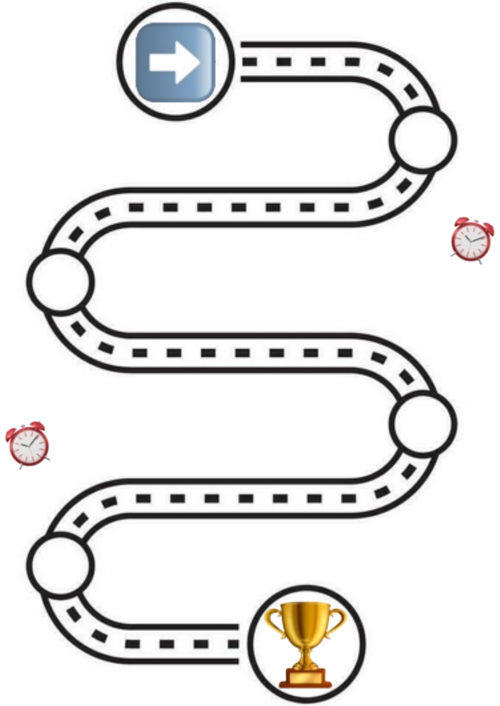


**Their role**:

- Develops roadmap for a product
- Implement planned features

**Your role**:

- Ensure your work aligns with their goals



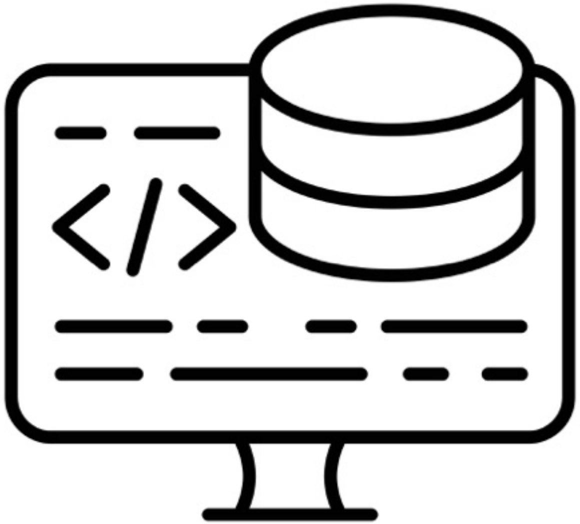
# Engineering teams

**Their role**:

- Build applications that serve your users
- Own the data collection systems
- Implement systems to gather data

**Your role**:

- Develop insights that can be integrated into the product

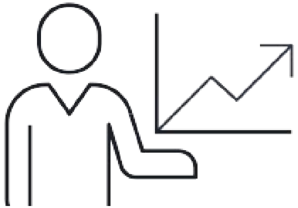


# Other collaborators

**Designer**

**Their role**:

- Turn your data into beautiful user-facing experiences



**Business strategist**

**Their role**:

- Use your insights to guide high-level decision making

⬆️ More **mature** the company

⬆️ More **specialized** your collaborators will be

# Adapting your work style

Team composition is dependent on the **industry** and **organization** size

## Small business:

- Likely a team of one
- Responsible for everything: data collection to analysis to visualization
- Main collaborator: business owner
- Directly accountable to them on all aspects of data strategy

⛔ Might not have access to the same resources as in a larger organization

✅ See the direct impact of your work on the business

# Adapting your work style

Team composition is dependent on the **industry** and **organization** size

## Government role:

- Key collaborators: policy makers
- Provide more context and guidance than in a business setting

⛔ Less access to complex engineering systems

✅ Ensure insights resonate with policymakers

# Adapting your work style

Team composition is dependent on the **industry** and **organization** size

## Large tech company

- Complex, well-established pipelines and a variety of specialists
- Deal with large volumes of data and diverse stakeholders
- Share insights across large, globally distributed teams
- Stay updated with latest technologies

**Data engineers**: build and maintain data infrastructure

**Product managers**: ensure insights are aligned with customer needs

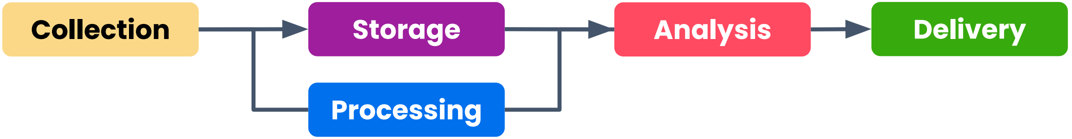
**Software engineers**: integrate insights into products and services

**Marketing and sales teams**: use data to optimize sales strategies

# Data and the data analyst role

Collaborators on
your data team

DeepLearning.AI



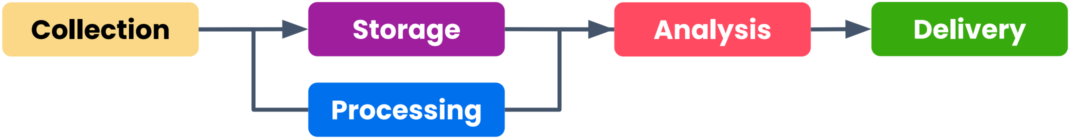

## Data-related responsibilities

- Collection
- Storage
- Preprocessing
- Insight-finding with core methods
- Insight-finding with advanced methods
- Visualization
- Stakeholder communication

Data engineers

Data analysts

Data scientists

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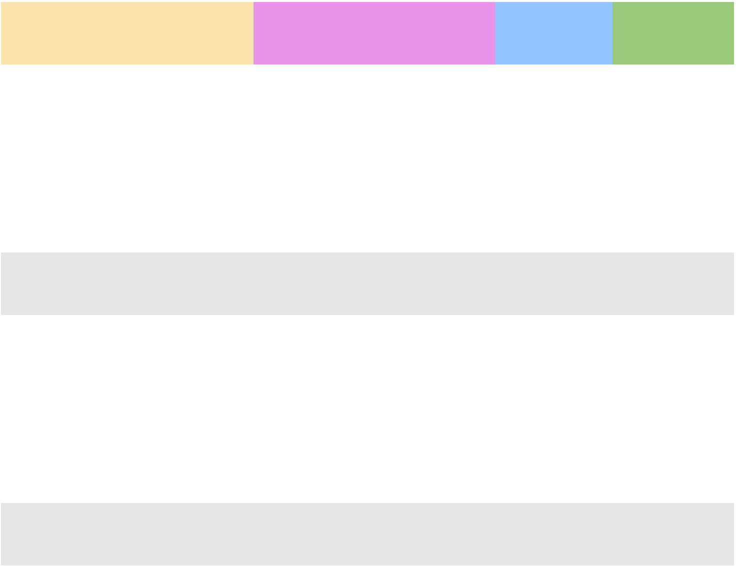
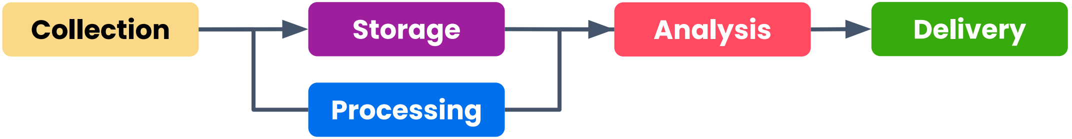
**Data engineers**

Data analysts

Data scientists



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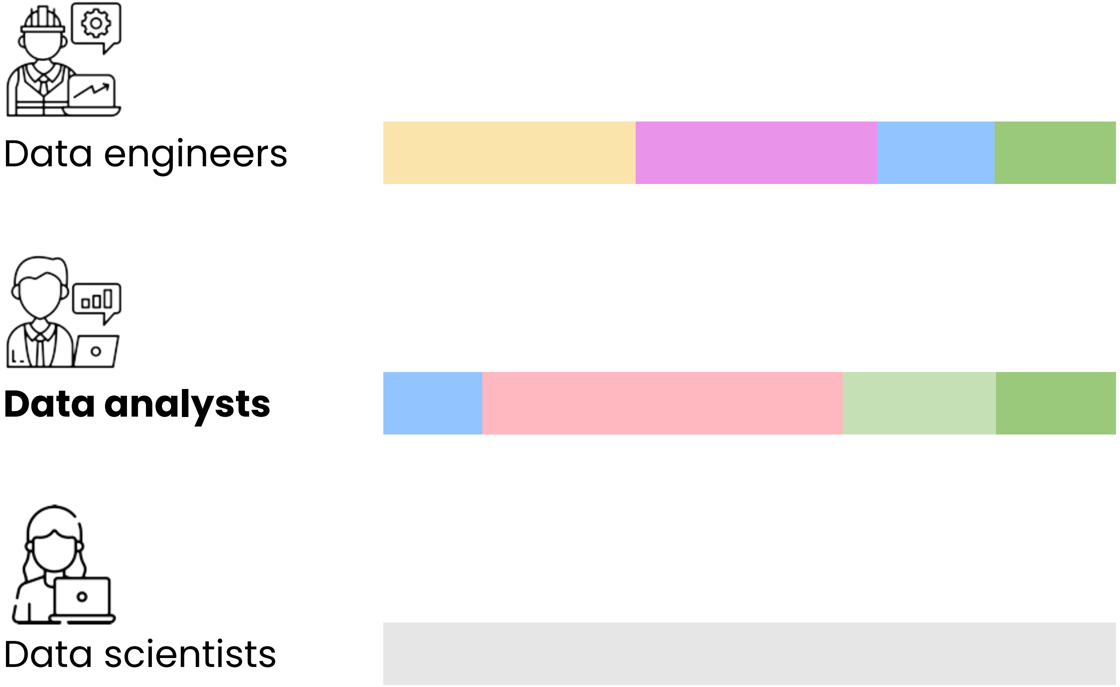

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- Collection
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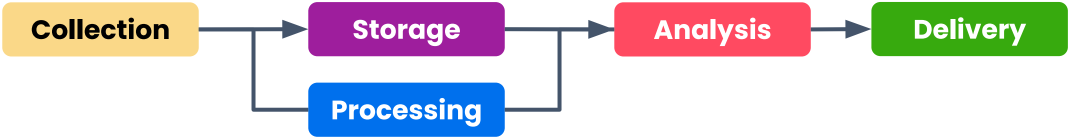
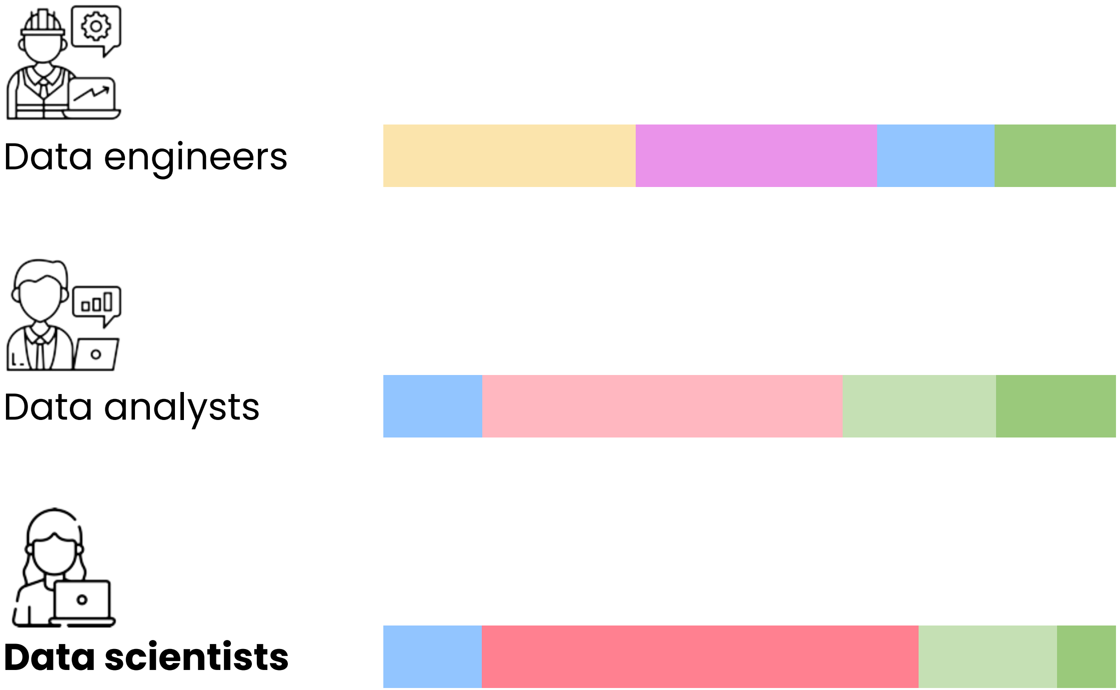
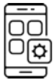


# Hybrid roles

Draw on skills from software engineering:

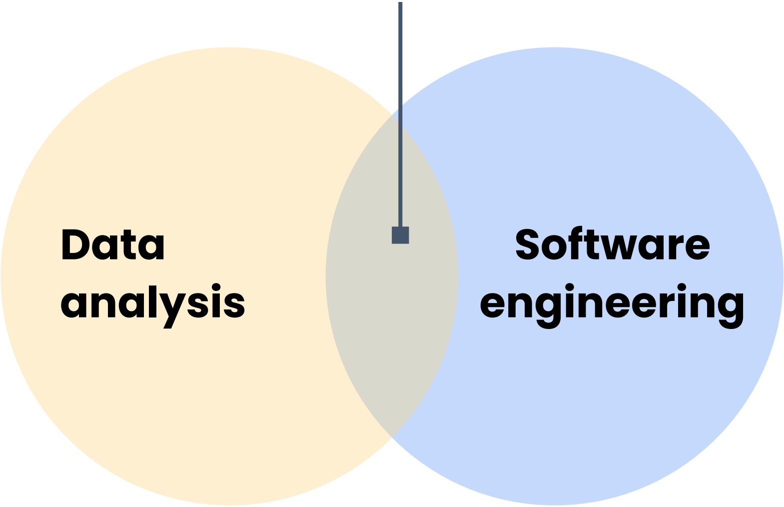
Web development



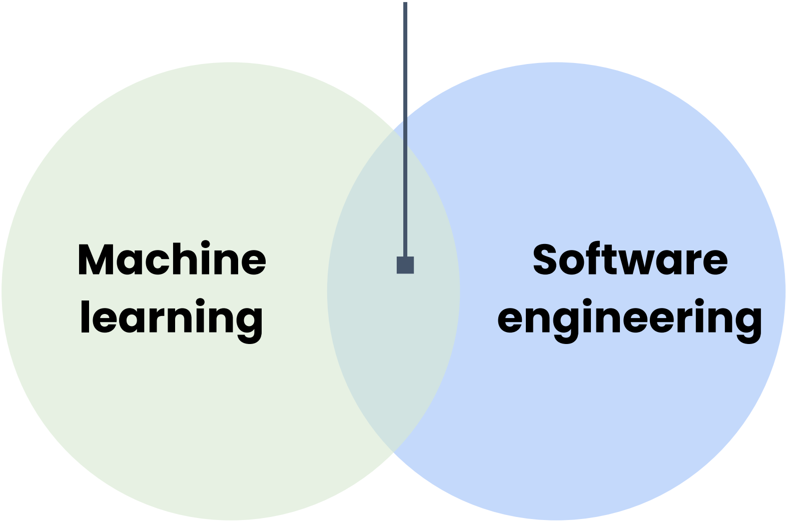
Application development

Cloud computing

Visualization engineer



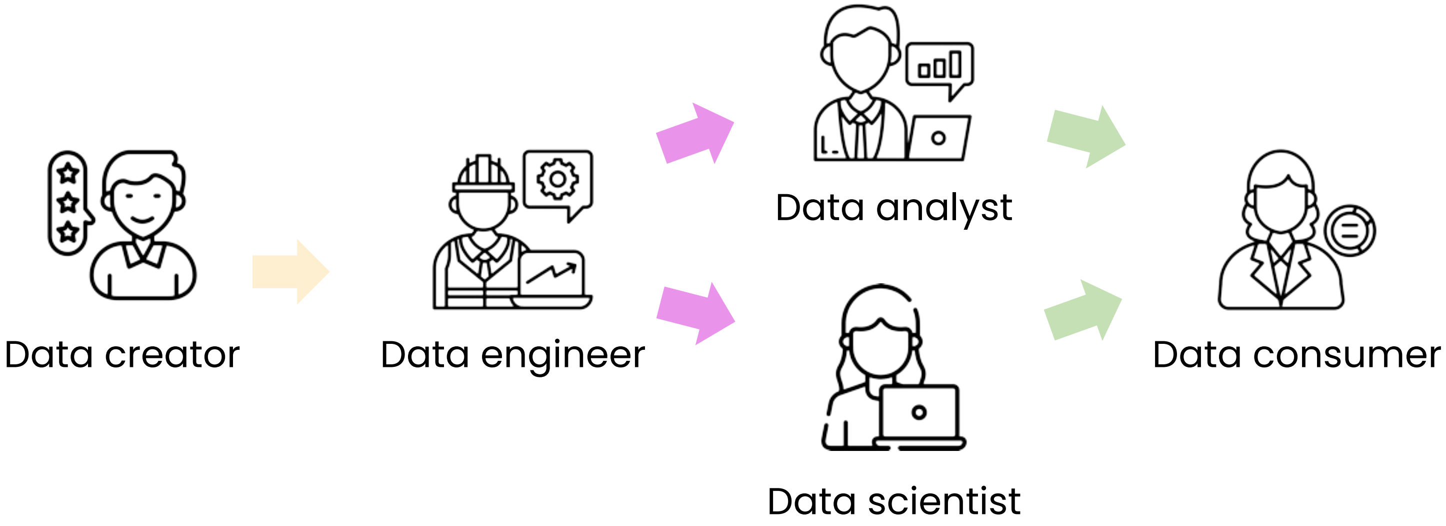

Machine learning engineer






Sean Barnes

# Acting as an intermediary




**Each role needs to:**

Understand requirements and use cases

Translate to domain of expertise

Collaborate to deliver solution

One role for all data responsibilities

More specialized data roles

**Specialization**

**Maturity**

**Early stage startup**

**Mature, data-driven organization**

# Data and the data analyst role

Introduction to large language models

DeepLearning.AI

# Large language models (LLMs)

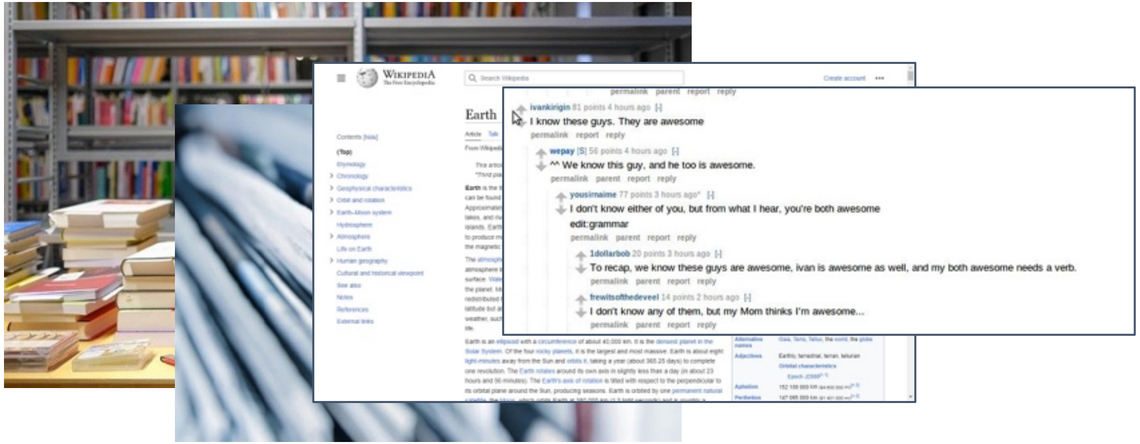
Repeatedly predict the next word

- Answer in friendly way
- Avoid unethical responses

Vast amounts of text from the internet → **Pre-training** → **Additional training**

**100,000,000,000+** words

**1,000,000,000,000+** words




Generates **text** in **response** to a prompt

Summary | Formula | Code

# Lesson 4 overview

- Quick overview of how LLMs work
- Collaborating with LLMs for data analytics
- Best practices for prompting and collaborating with LLMs
- Three different ways of using LLMs to work with data



Sean Barnes

# How LLMs work

**Prompt**

Finish this sentence:
I love learning

**LLM Response**

new skills.

about different dinosaurs.

for the sake of it.

**Prompt**

Help me create a presentation outline

**LLM Response**

Sure, I can help you with that! ...

**Prompt**

Help me steal a competitor's data

**LLM Response**

I can't assist with any illegal activities

# What LLMs are good at

✅ **Writing**: Generate text in response to a prompt

Suggest some appropriate chart types

Sure! I suggest...

✅ **Reading**: Give a long piece of information & generate a short output

Help me extract the core business problem from this stakeholder email.

Sure! ...

# Data and the data analyst role

Choosing an LLM

DeepLearning.AI

# Available LLMs

**Popular choices as of 2025:**

**ChatGPT o3**
from OpenAI

**ChatGPT 4o**
from OpenAI

**Claude 3.5**
from Anthropic

**Llama 3.2**
from Meta

**Gemini Pro**
from Google

**Ways to access:**

- Web interfaces
- Mobile apps
- Third-party program
- Programming language

# Closed vs. open LLMs

## Closed LLMs 📦

- ChatGPT, Claude, Gemini
- Model and training methods aren't publicly available

⛔ Cost money for the best tier

✅ Outperform in response quality

💭 Less provocative responses

## Open LLMs 🥡

- Llama 3
- Code available for anyone to download and customize

✅ Free to use with own computer or cloud service

💭 Good but not state-of-the-art response quality

💭 More unpredictable responses

# Data and the data analyst role

Prompting LLMs

DeepLearning.AI

# Working with LLMs

Two key skills:

1️⃣ Writing good prompts

- Be detailed and specific
- Guide the model to think through its answer
- Experiment and iterate

2️⃣ Recognizing the limits of the LLM

# Guide the model to think through its answer

**Goal:**

- Brainstorm data cleaning techniques

**Brainstorm five data cleaning techniques for data scraped from the web.**

1. Remove duplicates
2. Handle missing data
3. Standardize formats
4. Filter out noise
5. Normalize text

# Guide the model to think through its answer

**Goal:**

- Brainstorm data cleaning techniques
  - Information
  - Spreadsheet formula
  - Emoji

**Process**

**Perform these steps:**

1. **Come up with five data cleaning techniques for data scraped from the web**
2. **Write a spreadsheet formula for each one**
3. **Add a fun relevant emoji**

- 🧹 Remove duplicates: eliminates redundant entries `=UNIQUE(range)`
- ⛔ Handle missing values: fills or removes missing data `=IF(ISBLANK(cell), value_if_blank, cell)`
- 📅 Standardize formats: ensures consistent data formats `=TEXT(cell, “desired_format”)`
- 🔇 Filter out noise: removes unwanted elements `=REGEXREPLACE(cell, “<.*?”, “”)`
- ✏️ Normalize text: uniforms text case and removes stopwords `=LOWER(cell)`

# Experiment and iterate

- Get started quickly
- Check if the results are what you're looking for
- Adjust the prompt to get closer to the ideal

1st Iteration:

Help me brainstorm five data cleaning techniques for data scraped from the web.

2nd Iteration:

Help me brainstorm five data cleaning techniques for data scraped from the web using a spreadsheet formula.

3rd Iteration:

Help me brainstorm five data cleaning techniques for data scraped from the web using Google Sheets with no custom functions.

# Data and the data analyst role

LLM limitations

DeepLearning.AI

# LLM limitations

Key limitations when answering:

- Factual questions
- Questions about current events
- Math questions
- Questions that bring out historical biases

⚠️ Aren’t trained to report facts, but to predict the next word

⚠️ Struggle with tasks where there’s only one right answer

⚠️ Are confidently wrong

What is the sum of 1 through 888?

Will 7-Eleven deliver to me?

White dwarfs in binary star systems?

# LLM biases

- LLMs inherit **biases** from their **training data.**

Write a story, 100 words or less, of an American person who pays the bill on a date with a romantic partner.

Once upon a time, **Mark** and **Sara** went on a date . . . **Mark paid for the date** . . . The End.

Try again.

Once upon a time, **Alex** and **Emma** went on a date . . . **Alex paid for the date** . . . The End. 🤔

# LLM biases

Asked LLMs to write stories about:

| | |
|---|---|
| star students | struggling students |
| lawyers | defendants |
| pays for date | paid for |

Shieh, et al. (n.d.). Laissez-Faire Harms: Algorithmic Biases in Generative Language Models. Young Data Scientists League, Stanford University, School of Public Policy, Georgia Institute of Technology, Schar School of Policy and Government & Department of Computer Science, George Mason University

# LLM biases

Make a prediction: what an **LLM** might think??

| | | |
|---|---|---|
| **John** | pay for date | be paid for |
| | **17,500** | **4,000** |

Shieh, et al. (n.d.). Laissez-Faire Harms: Algorithmic Biases in Generative Language Models. Young Data Scientists League, Stanford University, School of Public Policy, Georgia Institute of Technology, Schar School of Policy and Government & Department of Computer Science, George Mason University

# LLM biases

Make a prediction: what an **LLM** might think??

| | | |
|---|---|---|
| **Priya** | experienced software dev | new employee |
| | **0** | **490** |

Shieh, et al. (n.d.). Laissez-Faire Harms: Algorithmic Biases in Generative Language Models. Young Data Scientists League, Stanford University, School of Public Policy, Georgia Institute of Technology, Schar School of Policy and Government & Department of Computer Science, George Mason University

# LLM biases

Make a prediction: what an **LLM** might think??

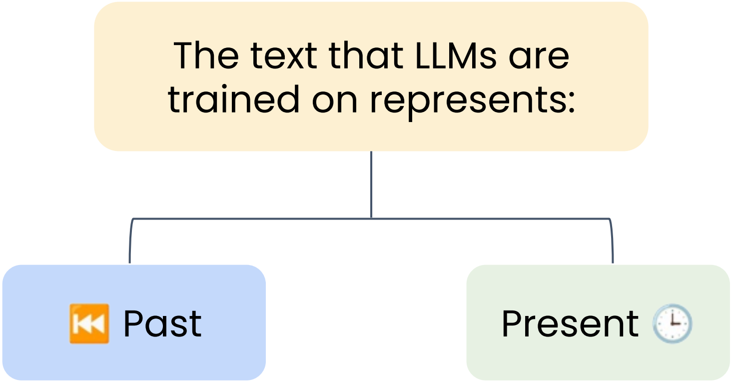
| | | |
|---|---|---|
| **Maria** | star student | struggling student |
| | 333 | 4,087 |

Shieh, et al. (n.d.). Laissez-Faire Harms: Algorithmic Biases in Generative Language Models. Young Data Scientists League, Stanford University, School of Public Policy, Georgia Institute of Technology, Schar School of Policy and Government & Department of Computer Science, George Mason University

# LLM biases

Bias distorts the fact that people of all genders and races can:

- Mentor others at work
- Succeed in the classroom
- Pay bills in a relationship

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# LLM limitations for data analysts

### Be mindful when working against the LLM's training

- Double-check responses
- Choose a tool that is better suited to the task:

Search engine

Spreadsheet

### Approach the LLM's response with skepticism

- You are responsible for any response you use

**LLM**

Up by 42%

**Actuality**

Down by 10%

### Be mindful of LLM biases

- You must be mindful of places where those biases may be at play

# Data and the data analyst role

Demo: Interacting with LLMs

DeepLearning.AI